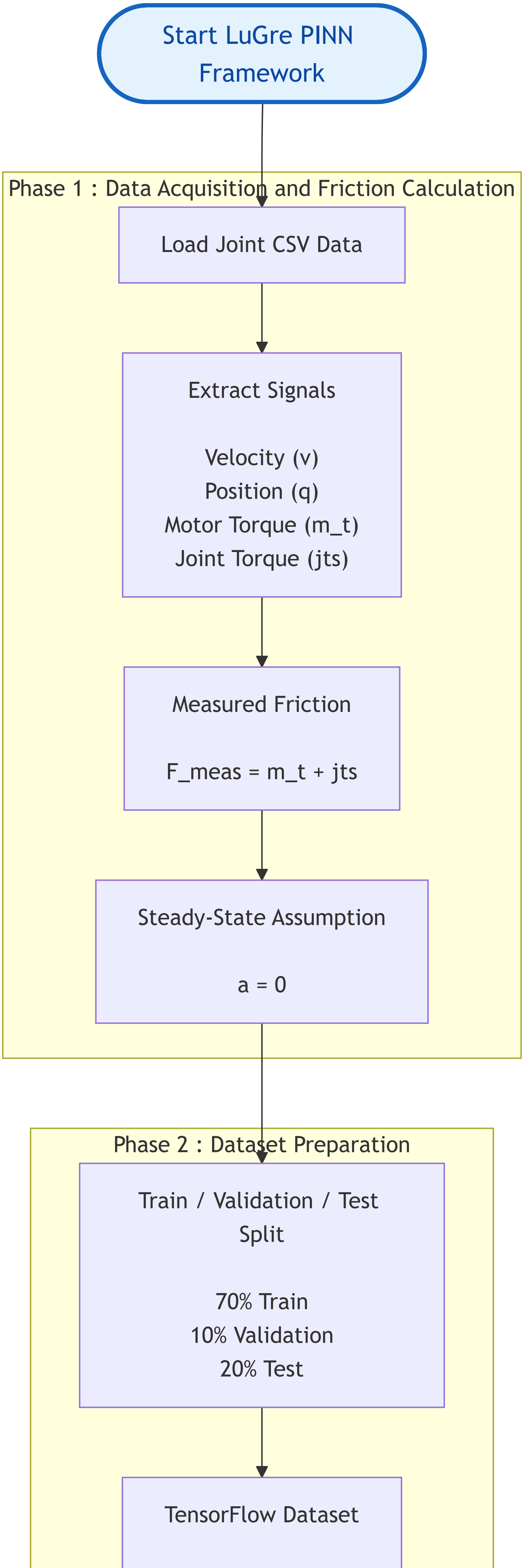
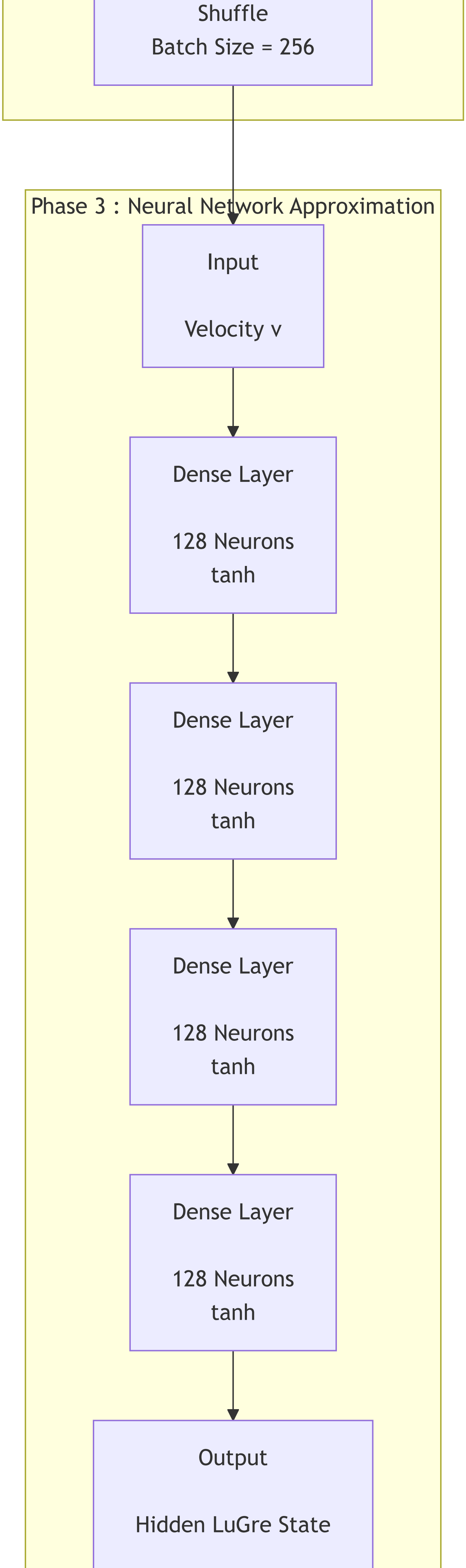
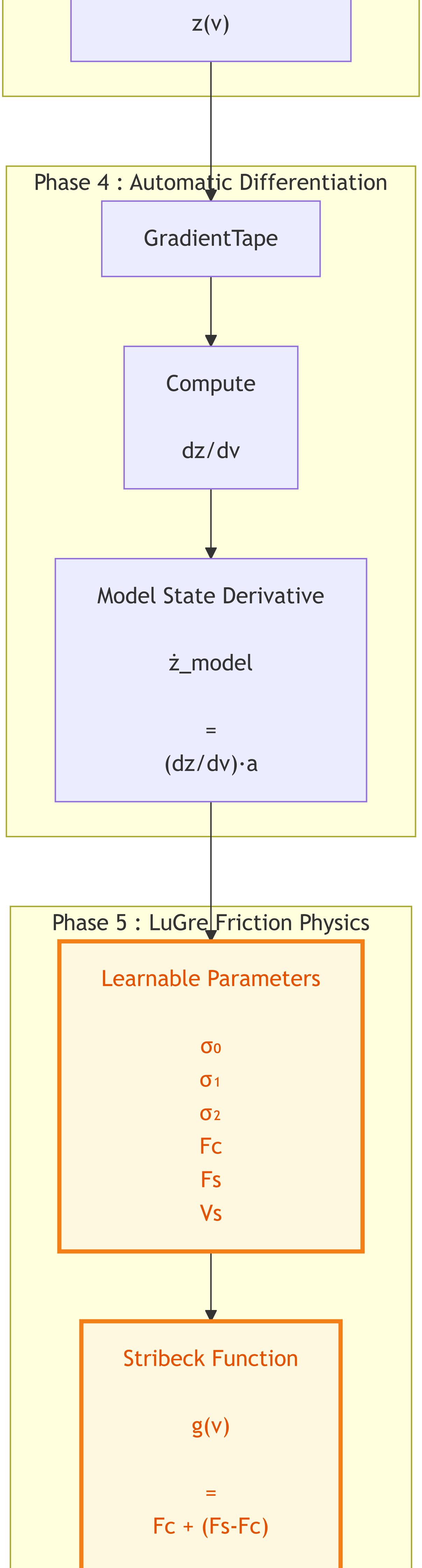
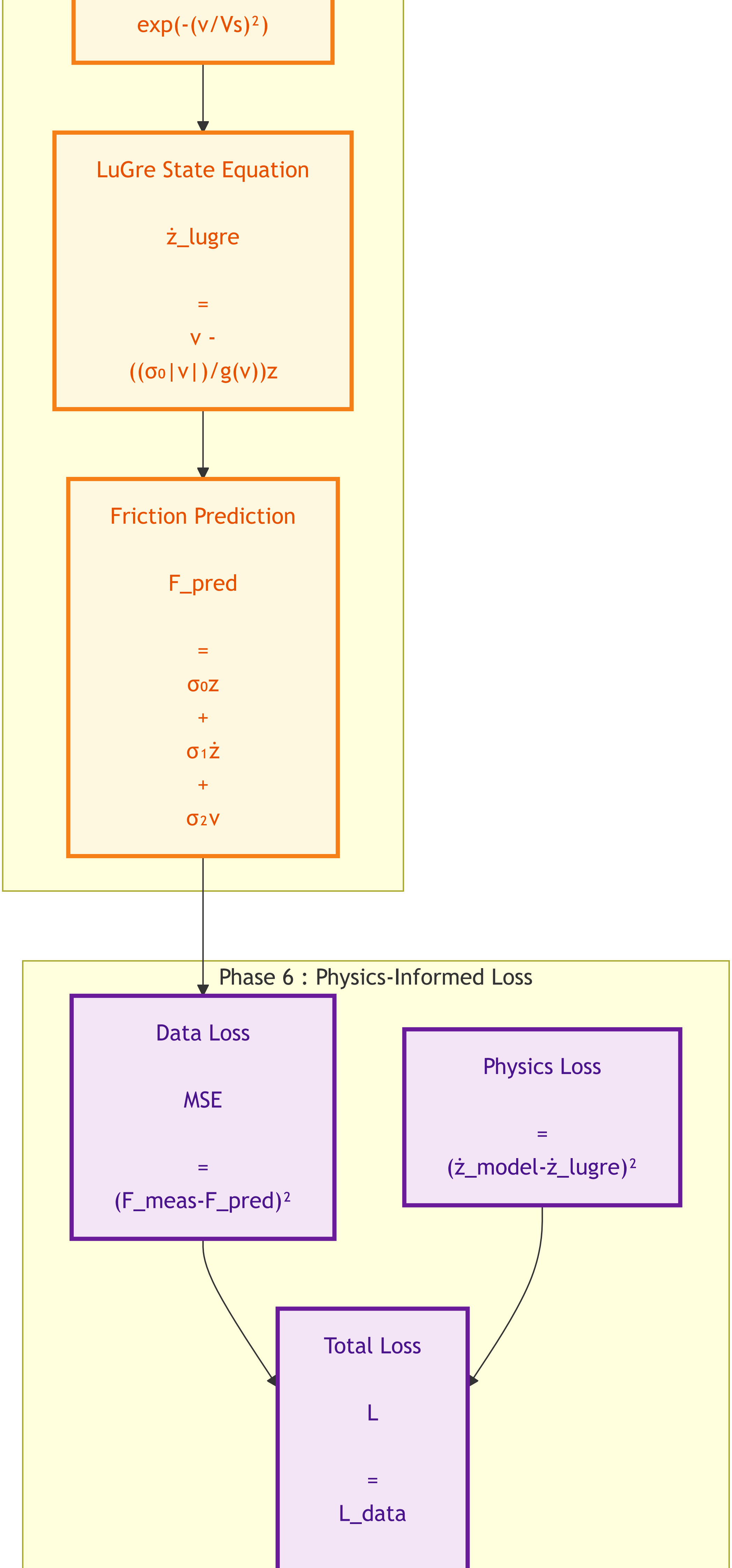
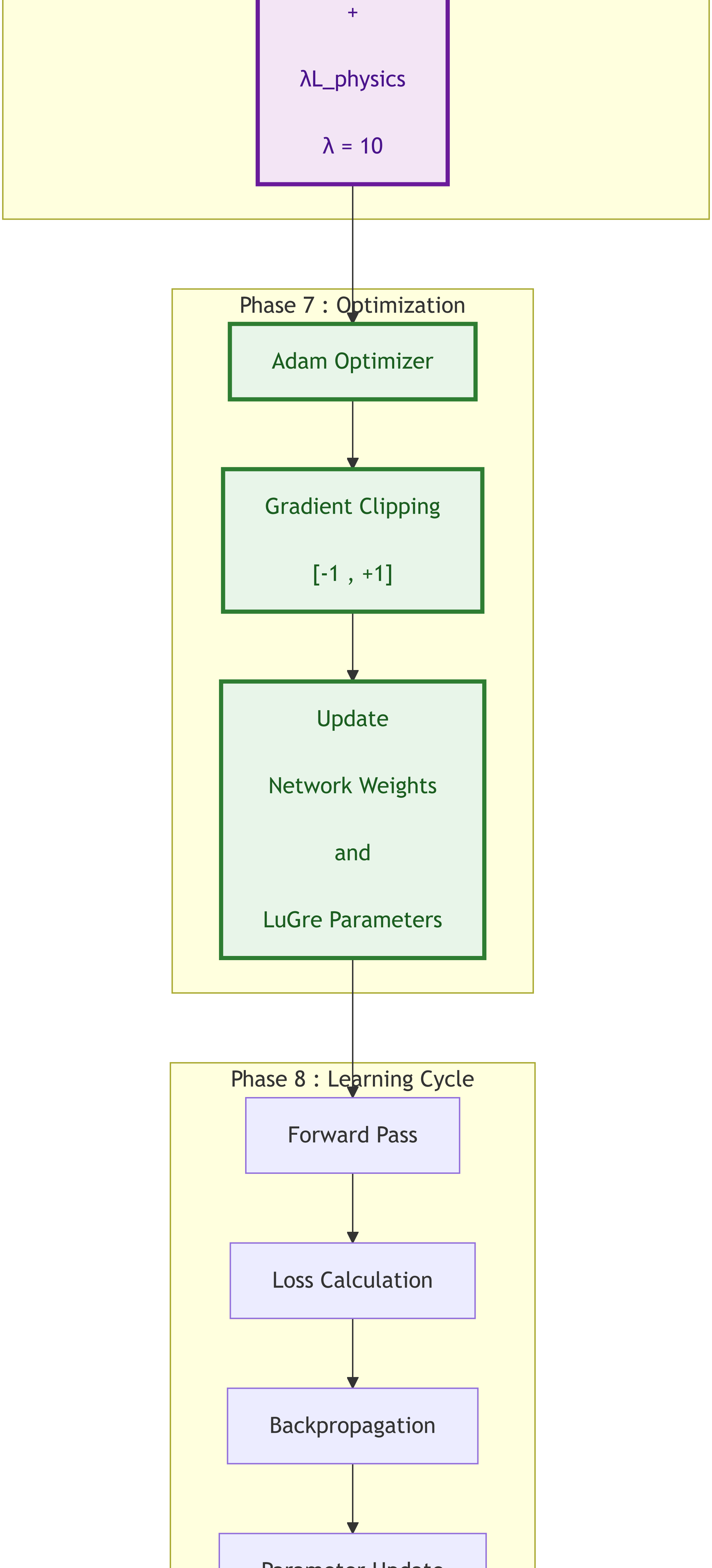

Start LuGre PINN Framework
Phase 1 : Data Acquisition and Friction Calculation
Load Joint CSV Data
Extract Signals
Velocity (v)
Position (q)
Motor Torque (m_t)
Joint Torque (jts)
Measured Friction
F_meas = m_t + jts
Steady-State Assumption
a = 0
Phase 2 : Dataset Preparation
Train / Validation / Test Split
70% Train
10% Validation
20% Test
TensorFlow Dataset


Shuffle
Batch Size = 256
Phase 3 : Neural Network Approximation
Input
Velocity v
Dense Layer
128 Neurons
tanh
Dense Layer
128 Neurons
tanh
Dense Layer
128 Neurons
tanh
Dense Layer
128 Neurons
tanh
Output
Hidden LuGre State

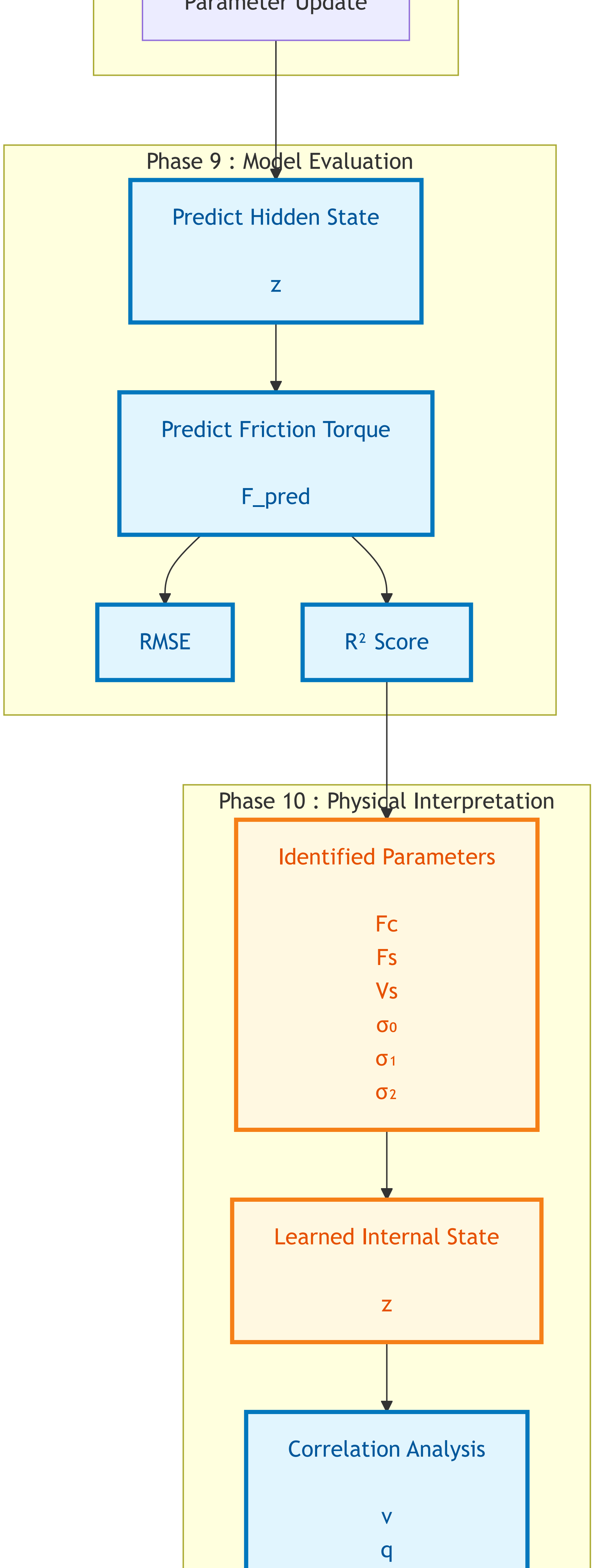


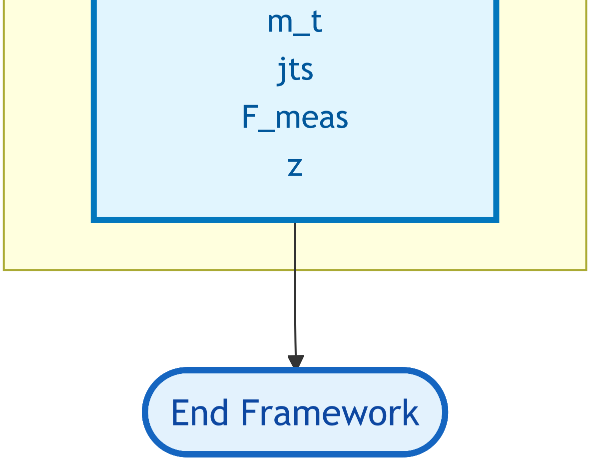
z(v)
Phase 4 : Automatic Differentiation
GradientTape
Compute
dz/dv
Model State Derivative
ż_model
=
(dz/dv)·a
Phase 5 : LuGre Friction Physics
Learnable Parameters
σ_0
σ_1
σ_2
Fc
Fs
Vs
Stribeck Function
g(v)
=
Fc + (Fs-Fc)


exp(-(v/Vs)²)
LuGre State Equation
ż_lugre
=
v -
((σ₀|v|)/g(v))z
Friction Prediction
F_pred
=
σ₀z
+
σ₁ż
+
σ₂v
Phase 6 : Physics-Informed Loss
Data Loss
MSE
=
(F_meas-F_pred)²
Physics Loss
=
(ż_model-ż_lugre)²
Total Loss
L
=
L_data



+
λL_physics
λ = 10
Phase 7 : Optimization
Adam Optimizer
Gradient Clipping
[-1 , +1]
Update
Network Weights
and
LuGre Parameters
Phase 8 : Learning Cycle
Forward Pass
Loss Calculation
Backpropagation


Parameter Update
Phase 9 : Model Evaluation
Predict Hidden State
z
Predict Friction Torque
F_pred
RMSE
R² Score
Phase 10 : Physical Interpretation
Identified Parameters
Fc
Fs
Vs
σ0
σ1
σ2
Learned Internal State
z
Correlation Analysis
v
q


m_t
jts
F_meas
z
End Framework